

O-RAN ALLIANCE Test and Integration Focus Group

Certification and Badging Processes and Procedures

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1	Contents	
2	Foreword	3
3	Modal verbs terminology	3
4	1 Scope	4
5	2 References	4
6	2.1 Normative references	4
7	2.2 Informative references.....	5
8	3 Definition of terms, symbols, and abbreviations.....	5
9	3.1 Terms 5	
10	3.2 Symbols.....	7
11	3.3 Abbreviations	7
12	4 Certification and Badging Overview.....	7
13	4.1 Objectives.....	7
14	4.2 Scope 8	
15	4.3 Entities Involved	10
16	4.3.1 O-RAN ALLIANCE	10
17	4.3.2 OTICs and the requesting entity(ies)	10
18	4.4 Outline of Processes and Technical Procedures.....	11
19	5. Certification and badging processes.....	12
20	5.1 General	12
21	5.1.1 Objectives of defining certification/badging processes	12
22	5.1.2 Principles.....	12
23	5.1.3 Overview	12
24	5.1.4 O-RAN Certificates and Badges	13
25	5.2 Processes	14
26	5.2.1 O-RAN Certification and Badging process flow	14
27	5.2.2 Defining Certification/Badging and Set of Tests.....	16
28	5.2.3 Test platforms.....	16
29	5.2.4 Identifying Applicable O-RAN certificate/badge and Tests.....	16
30	5.2.5 Executing Tests	17
31	5.2.6 Verifying Test Results.....	17
32	5.2.7 Issuing Certificates/Badges.....	17
33	5.2.7.1 Certificate/Badge Reference Identifier.....	17
34	5.2.8 Compliance Folder	18
35	5.2.9 Storing and Sharing of Summary Reports, Test reports and Measurement Results	18
36	5.2.10 SH/HW changes to certified/badged devices	19
37	5.2.11 Multi-vendor Scenario.....	19
38	5.3 Defined and Available O-RAN Certificates and Badges	20
39	6. O-RAN Certificate Procedures.....	21
40	6.1 Subsystems and Interfaces	21
41	6.2 O-RAN Open Fronthaul interface of O-RU.....	21
42	6.3 O-RAN Open Fronthaul interface of O-DU.....	22
43	6.4 O-RAN A1 services on A1 interface of Non-RT RIC	22
44	6.5 O-RAN A1 services on A1 interface of Near-RT RIC	23
45	7. O-RAN IOT Badge Procedures.....	23
46	7.1 Subsystems and Interfaces	23

1	7.2 O-RAN Open Fronthaul interface.....	24
2	7.3 X2 interface.....	24
3	7.4 F1 interface.....	25
4	7.5 Xn interface.....	25
5	7.6 A1 services on A1 interface.....	25
6	8. O-RAN E2E Badge Procedures.....	26
7	Annex A (informative): Certification application.....	27
8	Annex B (informative): Compliance folder.....	28
9	Annex C (normative): Trademark License Agreement.....	29
10	Annex D (normative): Machine readable results artifact.....	33
11	Annex (informative): Change history/Change request (history).....	38

12 Foreword

13 This Technical Specification (TS) has been produced by TIFG (Test and Integration Focus Group) of the O-
14 RAN ALLIANCE.

15 The content of the present document is subject to continuing work within O-RAN and may change following
16 formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will
17 be re-released by O-RAN with an identifying change of version date and an increase in version number as
18 follows:

19 version xx.yy.zz

20 where:

- 21 xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements,
22 corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with
23 leading zero if needed.
- 24 yy: the second digit-group is incremented when editorial only changes have been incorporated in the
25 document. Always 2 digits with leading zero if needed.
- 26 zz: the third digit-group included only in working versions of the document indicating incremental
27 changes during the editing process. External versions never include the third digit-group. Always 2
28 digits with leading zero if needed.

29

30 Modal verbs terminology

31 In the present document "shall", "shall not", "should", "should not", "may", "need not", "will", "will not",
32 "can" and "cannot" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal
33 forms for the expression of provisions).

34 "must" and "must not" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

35

1 Scope

The present document describes the certification and badging processes and associated detail technical procedures that Open Testing and Integration Centre (OTIC) [1] and other testing entities should adopt to:

- 1) Verify compliance of devices under test (DUTs) using O-RAN interface or an implementation of O-RAN reference design based on conformance test specifications (**conformance certification**),
- 2) Assess interoperability of DUT pairs using O-RAN interfaces based on Interoperability Test (IOT) specifications (**IOT badging**),
- 3) Assess end-to-end system integration of SUT using O-RAN interfaces based on E2E test specifications (**E2E badging**).

The present main document consists of the following subsidiary documents:

- O-RAN.TIFG.TS.Cert-Badge.1 contains O-RAN Certificate and Badge Summary Reports
- O-RAN.TIFG.TS.Cert-Badge.2 contains template of O-RAN Certificate and Badge Test Report
- O-RAN.TIFG.TS.Cert-Badge.3 contains JSON scheme for storing machine-readable test results
- O-RAN.TIFG.TS.Cert-Badge.4 contains O-RAN Certification and Banding Profiles (CB-Profiles)
- O-RAN.TIFG.TS.Cert-Badge.5 contains O-RAN Certification and Badging ICS Proformas (CB-ICS Proformas)

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.TIFG.CGofOTIC.0-v05.00: O-RAN ALLIANCE Test and Integration Focus Group, Criteria and Guidelines of Open Testing and Integration Centre, Version 05.00, October 2022
- [2] O-RAN.TIFG.E2E-TST.0-v04.00: O-RAN ALLIANCE Test and Integration Focus Group, End-to-end Test Specification, Version 5.0, February 2024

- [3] O-RAN.WG1.OAD-R003-v10.00: O-RAN ALLIANCE Working Group 1, O-RAN Architecture Description, Version 10.0, October 2023
- [4] 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2"
- [5] 3GPP TS 38.401: "NG-RAN; Architecture description"
- [6] 3GPP TS 37.340: "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity; Stage 2"

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

3 Definition of terms, symbols, and abbreviations

3.1 Terms

For the purposes of the present document, the terms [given in [i.1] and the following] apply:

Badging: the process by which interoperability for a pair of products or E2E system integration for a group of products are assessed, and the involved products are authorized to carry a specific set of credentials that indicate successful results. Badging information, which is common to all products involved, will include information on all products (including hardware version and software version) which were involved in interoperability testing or E2E system testing.

Blueprint: the set of inputs which is used to describe a specific deployment of O-RAN-based system or solution from several aspects. These aspects include the specification of the O-RAN deployment at the system level (e.g. architecture, performance metrics) as well as subsystem level and the interfaces between the specified subsystems. This would then allow definition and documentation of the testing methodology and the test cases using the blueprint specification.

Catalogue: the centralized database providing basic (non-sensitive) information about all issued certificates, IOT badges and E2E badges. This catalogue is publicly available on the O-RAN ALLIANCE web site and maintained by O-RAN ALLIANCE.

Certification: the process by which compliance with O-RAN interface or reference design specifications for a product is verified, and the product is authorized to carry a specific set of credentials that indicate that it is conformant.

Compliance folder: the set of detailed measurement results demonstrating an O-RAN ALLIANCE device's compliance with the Badging and Certification Criteria, which includes the ICS, test reports, summary reports, technical documentation, change management documentation and certification information.

Implementation Conformance Statement (ICS): statement of conformance provided by requesting entity(ies) who bring their DUTs/SUT for badging and/or certification service, which is the key element in determining which test cases will be executed on a particular product.

Logo: O-RAN approved logo for certification and badging differentiating products/solutions that passed certification, IOT badging, and/or E2E badging. The logo is a part of O-RAN certificate, IOT badge or E2E badge. The logo is intended to be utilized for marketing purposes at website, brochure, printed at the product itself, etc.

Measurement results: a collection of detailed measurement results collected during the certification or badging including logs, screenshots, packet captures, etc. The file format of measurement results is not specified and unified and depends on used test and measurement equipment and tools.

MVP: the minimum viable plan to be tested for Certification and Badging.

Test report (O-RAN Certificate and Badge Test Report): provides detailed information about DUT/SUT, test setup configuration, assessment of measured KPIs, gap analysis, evidence of meeting pass criteria, etc. The unified format of document (template) is used to ensure reusability and sharing of the information. The test report is a part of O-RAN certificate, IOT badge or E2E badge.

Summary report (O-RAN Certificate and Badge Summary Report): provides a summary of O-RAN certification, IOT badging or E2E badging testing including basic information about issued O-RAN certificate, IOT badge or E2E badge, contacts to all involved entities, list of awarded DUTs, list of used test and measurement equipment and tools, list of applicable and conducted tests, etc. The unified format of document (template) is used to ensure reusability and sharing of the information. The Summary report is a part of O-RAN certificate, IOT Badge and E2E badge.

Certification and Badging ICS Proforma (CB-ICS Proforma): An overview of all mandatory/optional/conditional functions and capabilities defined in base specification(s)/profile and, if exists, in O-RAN Certification and Badging Profile. It helps vendors ensure that their products meet the necessary requirements for O-RAN Certificates and Badges. When the implementation is done, vendor declares which functions and capabilities have been implemented/supported, and so it becomes Certification and Badging ICS.

Certification and Badging ICS (CB-ICS): A statement made by the vendor of an implementation stating which functions and capabilities have been implemented/supported for the purpose of O-RAN Certificate or Badge. The Certification and Badging ICS (CB-ICS) is attached to O-RAN Certificate, IOT Badge or E2E Badge.

Certification and Badging Profile (CB-Profile): For the purpose of O-RAN Certification and Badging, the CB-Profile may constrain the existing options in base specification(s)/profile, and shall not impose requirements that conflict with the existing requirements in base specification(s)/profile (e.g. a function/capability/parameter mandatory in the base specification shall not be optional in the profile; or adding new function/capability/parameter that does not exist in the base specification is not allowed).

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

CB-ICS	Certification and Badging Implementation Conformance Statement
CB-Profile	Certification and Badging Profile
DUT	Device Under Test
E2E	End-to-End
FFS	For Future Study
ICS	Implementation Conformance Statement
IOT	Interoperability Testing
IUT	Interface under Test
MVP	Minimum Viable Plan
Near-RT RIC	Near-Real Time RAN Intelligent Controller
Non-RT RIC	Non-Real Time RAN Intelligent Controller
O-CU	O-RAN Central Unit
O-DU	O-RAN Distributed Unit
O-RU	O-RAN Radio Unit
OTIC	Open Testing and Integration Centre
PTC	Profile Test Configuration
RIC	RAN Intelligent Controller
SMO	Service Management and Orchestration
SUT	System Under Test
TIFG	O-RAN Test and Integration Focus Group

4 Certification and Badging Overview

4.1 Objectives

The motivation of O-RAN certification and badging is to minimize the repetition of fundamental and common tests which should be performed to verify and validate O-RAN compliant products, systems, and solutions before their deployment in operator networks thus reduce the test effort. The idea is to test only once and reuse/adopt the test results multiple times whenever it is possible. In addition, the use of O-RAN certificates and badges for network components can be a very important way to promote the O-RAN ecosystem and to encourage implementation and testing of O-RAN products, systems and solutions and their deployment in operator networks.

The purpose of **O-RAN certification** is to validate and certify that a vendor product is compliant to a related set of O-RAN defined interface or reference design specifications through a set of O-RAN defined conformance tests using unified test procedures and templates for sharing of results. The O-RAN certificate is awarded to a product for one or more interfaces (e.g. O-RU CUSM-OpenFH, O-RU O1, O-DU O1, O-cloud O2).

The purpose of **O-RAN IOT badging** is to prove interoperability of a pair of products based on the most fundamental and operator neutral (common) features and parameters through a set of O-RAN defined interoperability tests using unified test procedures and templates for sharing of results. In addition, O-RAN IOT badging can also provide an indication of interoperability of a pair of products based on optional/advanced and operator specific features and parameters through a set of O-RAN defined interoperability tests using unified test procedures and templates for sharing of results. The O-RAN IOT badge is awarded to both products in the DUT pair and is applicable for an O-RAN interface.

The purpose of **O-RAN end-to-end badging** is to demonstrate and verify that a (multi-vendor) end-to-end system meets minimum requirements in terms of functionality, security and other mandatory features defined in related O-RAN test specifications. In addition, O-RAN end-to-end badging can also provide an indication of end-to-end performance and other optional end-to-end capabilities through a set of O-RAN defined end-to-end tests using unified test procedures and templates and data formats for sharing of results. The O-RAN E2E badge is awarded to all products in the SUT and is applicable to the specific scope tested.

With certification and badging in place, operator awareness of a vendor O-RAN-based system or solution is expected to increase, and the operator own testing is expected to reduce because the common tests can be performed only once and would not need to be repeated.

O-RAN certification and badging may also be helpful for vendors as it provides some level of evidence that their O-RAN-based systems and solutions work, which could be used to promote their systems and solutions to operators. Further, for vendors looking for other vendors to interoperate with, certification and badging may also be beneficial to increase acceptance from other vendors.

O-RAN certification and badging should not be used as a substitute for operator-specific testing of O-RAN based systems and solutions prior to deployment in operator networks. O-RAN certificate and badge do not mean all the necessary tests have been performed and passed, and O-RAN certified/badged system or solution can be automatically deployed in an operator network without any additional testing. Rather, O-RAN certification and badging can improve operator confidence in their chosen O-RAN based blueprint and can reduce the complexity and duration of pre-deployment testing.

4.2 Scope

The document defines and unifies the processes, procedures, templates, data format, etc. to ensure reusability and sharing of the test results and repeatability of the executed tests. The required tests for issuing of specific O-RAN certificate/badge are decoupled from this document. The document only references O-RAN test specification(s) (but without mentioning a specific version) which shall be used for issuing of specific O-RAN certificate/badge. All the test related information such as test description, test applicability, test procedure, test setup and configuration, expected results, recorded KPIs, pass/fail criteria are defined in the O-RAN test specifications and shall be properly followed. If needed, the specific modifications, clarifications and alignments of O-RAN test specifications required only for the purpose of O-RAN certificates and badges are defined in this document. It is also assumed that the user/tester has properly read and understood the related O-RAN test specifications. In case of any unclarity in O-RAN test specification, specific O-RAN WG/FG owning O-RAN test specification shall be contacted to resolve the unclarity.

For certification and badging purposes, OTICs and requesting entity(ies) should use the latest published version of this document. The summary report and test report shall include the versions of this document and relevant O-RAN test specification used to perform O-RAN certification or badging. Each O-RAN certificate or badge shall only be issued against a single version of the corresponding test specification which is indicated

in the summary report and test report (i.e. a certificate/badge cannot be issued based on mixing tests from multiple test specification versions).

The certification and badging described in this document defines the processes and associated technical procedures in the following areas:

- Compliance verification of DUTs using O-RAN interface or an implementation of O-RAN reference design based on conformance test specifications (**conformance certification**)
- Interoperability assessment of DUT pairs using O-RAN interfaces based on IOT test specifications (**IOT badging**)
- E2E system integration assessment of SUT using O-RAN interfaces based on E2E test specifications (**E2E badging**)

The reason to use the term “certification” for conformance testing and “badging” for interoperability and E2E system testing is as follows:

- Conformance tests are controllable as it only involves a single DUT and verifies whether that DUT behaves in accordance with a technical specification (e.g. open fronthaul interface specification);
- IOT and E2E system tests have some variability as it involves multiple DUTs/SUT (from different vendors) and the results may depend on the DUT/SUT combinations, deployment blueprints and test profiles.

Figure 4.2-1 provides an illustration for the above-described concept of O-RAN certification and badging.

Objectives	Compliance Verification and Interoperability & E2E Use Case/System Assessment				
Technical Specifications	Technical Specifications (normative)				
	Open Interfaces / APIs	Open Subsystems	O-RAN Use Cases ¹	Open Subsystems & Interfaces / APIs	E2E System
Assess implementation	Compliance Verification		Interoperability and E2E Assessment		
Test specs	Conformance Test Specifications (WGs)		Use Cases Test Spec (WGs) ¹	IOT Specifications (WGs)	E2E System Testing Specifications (TIFG)
Test objects	Single DUT and Interface/API	Subsystem—whitebox, O-cloud, stacks	SUT inc. RICs & I/Fs, APIs ¹	1 Pair of DUTs with a single Interface / API	SUT with More than 2 DUTs & I/Fs, APIs
Testing Outcomes	Certification		Certification or Badging ¹	Badging	

¹ this is still to be discussed within TIFG

Figure 4.2-1: O-RAN certification and badging concept

Whether or not and how to address 3GPP functional and performance testing for the underlying components and interfaces is for future study (FFS).

4.3 Entities Involved

4.3.1 O-RAN ALLIANCE

In terms of certification and badging, O-RAN Test and Integration Focus Group (TIFG) only specifies the processes and associated detailed technical procedures, identifies the relevant O-RAN test specifications and test cases specified by O-RAN WGs/FGs, including end-to-end tests defined by O-RAN TIFG [2].

4.3.2 OTICs and the requesting entity(ies)

O-RAN certification and badging can only be performed in qualified O-RAN Open Testing and Integration Centers (OTICs), and consequently the O-RAN Certificates and Badges (with O-RAN approved logos for certification and badging) can only be issued by OTICs.

Under no circumstances are OTICs under the charter or governed/sponsored by O-RAN/TIFG. It is OTICs that will take full responsibility on all certification/badging related activities including certification/badging implementation, verification/assessment as well as certificate/badge issuance.

O-RAN Certificates and Badges (with O-RAN approved logos for certification and badging) can be awarded to a product provided by any requesting entity with or without O-RAN membership (O-RAN Member, O-RAN Contributor, O-RAN Academic Contributor). OTICs cannot be the requesting entity for its own certification/badging.

Further, the requesting entity can be the product original equipment vendors/manufacturers (OEMs), or product owners that have acquired and owned the products. Upon the completion of Certification/Badging and then the award of O-RAN Certificate/badge, the requesting entity is the copyright co-holder of summary and test reports, and can use Certificate/Badge logo for its marketing,

In case the requesting entity is the product owner but not the original equipment vendor/manufacture, the following extra steps shall be followed:

- The product owner should inform the original equipment vendor/manufacture of the certification/badging request even though the consent of such request from the original product vendor/manufacture is not required,
- The product owner does not need to share the test results/reports with the original product vendor/manufacture,
- The original product vendor/manufacture cannot use Certificate/Badge logo for its marketing without the consent of the product owner who requests for the O-RAN certification/badging,

In case an O-RAN Certificate or Badge is awarded to a product provided by a requesting entity without O-RAN membership, the legally binding "Trademark License Agreement" (see Annex C) between O-RAN ALLIANCE and that requesting entity without O-RAN membership shall be signed first. The "Trademark License Agreement" is signed only once when the first O-RAN Certificate or Badge is awarded. The list of entities with signed "Trademark License Agreement" is maintained on [O-RAN wiki](#), and shall be checked by the OTIC before awarding an O-RAN Certificate or Badge to any entity without O-RAN membership. If an entity is not listed, the entity shall contact O-RAN ALLIANCE (directly or through OTIC) to sign the "Trademark License Agreement".

In case an O-RAN Certificate or Badge is awarded to a product provided by an entity with O-RAN membership, no additional agreement is needed. The trademark license policies are part of O-RAN membership agreement signed by each O-RAN Entity. The list of entities with O-RAN membership is available on the [O-RAN wiki](#).

OTICs provide a collaborative, open, and impartial working environment, and the criteria and guidelines for OTICs are specified in [1]. These guidelines stipulate that:

- O-RAN summary reports, test reports and detailed measurement results can be freely accessed by O-RAN operator members for the purpose of evaluating the equipment for use in their networks
- An OTIC may or may not choose to provide certification and/or badging services. If an OTIC intends to provide O-RAN certification/badging services, it is mandatory for them to follow the processes and associated technical procedures specified in this document.
- Prerequisites for OTICs to be able to provide certification/badging services are defined in [1].

O-RAN allows the testing to be also performed outside of the OTICs using O-RAN test specifications and following the certification and badging procedures as defined in this document. However, such test results cannot carry an O-RAN Certificate or O-RAN Badge with O-RAN approved logos but may state that their equipment has been "tested according to O-RAN Certification and Badging Process and Procedures document."

OTICs may require certain pre-requisite testing to be done by the requesting entity(ies) before the start of the certification/badging process. Some examples are illustrated as follows

- 3GPP compliance validation as a prerequisite for any O-RAN conformance certification
- O-RAN conformance certification as a prerequisite for O-RAN IOT / E2E badging

4.4 Outline of Processes and Technical Procedures

Certification and badging include various processes and technical procedures.

Chapter 5 will define these processes in following areas:

- Describe the Certification and Badging Profile and ICS Proforma/ICS
- Describe the process of verifying certification/badging test results and responsible entity
- Describe how to issue a certificate/badge
- Describe how to share O-RAN summary reports, test reports and detailed measurement results
- Describe how to improve/enhance the certification/badging process and what triggers a new release of this document (e.g. WG spec 1.0 to 2.0, O-RAN Software Community Bronze release to Cherry)

Corresponding to processes to be described in Chapter 5, the detail technical procedures introduced in the following Chapters include identifying all relevant test specifications given by various O-RAN WGs/FGs in terms of Conformance, IOT, and E2E validation of underlying subsystems and interfaces.

Chapter 6 defines the procedures related to Conformance certification that must be performed on the DUT.

Chapter 7 defines the procedures related to the IOT badging that must be performed on the DUTs.

Chapter 8 defines the procedures related to E2E badging that are applicable for the operator deployment scenario (e.g. LTE, 5G NSA, or 5G SA). This can also include some unique O-RAN features or use cases that are part of O-RAN MVP initiative.

5. Certification and badging processes

5.1 General

5.1.1 Objectives of defining certification/badging processes

The objective is to have in place some level of uniformity on the certification/badging processes in order to:

- Ensure quality of the certification/badging service provided by different OTICs.
- Align procedures that users of the certification/badging service (both requesting entity(ies) bringing in DUT(s)/SUT(s) for certification/badging, and operators requesting access to summary reports, test reports and detailed measurement results of certified/badged DUTs/SUT) experience at different OTICs.

5.1.2 Principles

The certification/badging process sub-chapter adheres to the following principles:

- A distributed approach is adopted in which (a) roles pertaining to certification/badging are allocated to OTICs and (b) related details are left to OTICs' discretion and responsibility. Rather than having rigorous centralized O-RAN control and management, this approach helps avoid imposing heavy requirements and allows freedom of operation to OTICs. However, a centralized approach will also be adopted where required, e.g. to ensure quality of certification/badging, sharing of summary reports, test reports and detailed measurement results, etc.
- Defining certification/badging process elements to address near-term needs of the community, while also establishing a framework that can evolve and scale with the ecosystem growth, i.e. this sub-chapter can evolve over time.
- O-RAN will, as necessary, issue new versions of its functional specifications. As a result, O-RAN may determine that it will launch an updated version of its certification/badging program.
- O-RAN will, as necessary, determine that an existing version of its functional specification is no longer viable and should be taken out of service for product development and certification/badging. As a result, O-RAN may determine when the specification is to be taken out of service.

5.1.3 Overview

Figure 5.1-1 shows the relation between O-RAN and OTICs.

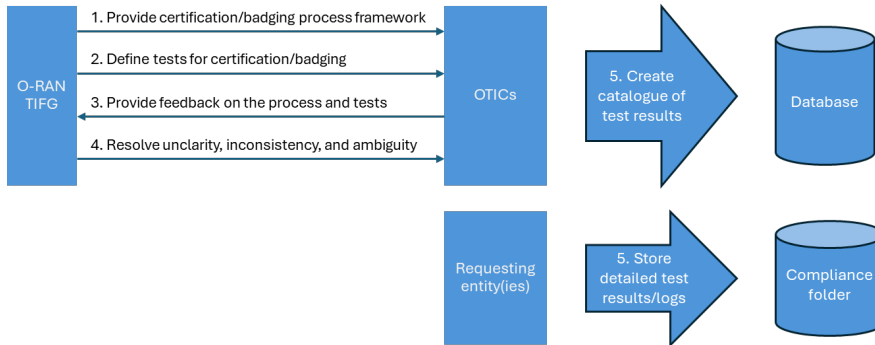


Figure 5.1-1: Relation between O-RAN and OTICs

1. O-RAN provides the certification and badging process framework in this document. OTICs should follow the processes described in this document.
2. O-RAN defines tests for certification/badging. Further description is provided in Section 5.2.1.
3. OTICs can provide feedback to O-RAN on the certification/badging process and tests for certification/badging, making it a closed loop operation.
4. O-RAN TIFG is the governing body to resolve any un-clarity, inconsistency, and ambiguity in the present document.
5. OTICs create a catalogue or database to share certification test reports with the vendors whose subsystems/systems were tested, operators, and other OTICs. Detailed measurement results should be stored by the requesting entity(ies) themselves as evidence. Further description on sharing of summary reports, test reports and detailed measurement results is provided in Section 5.2.8 and on SW/HW update in Section 5.2.9.

Figure 5.1-2 shows the certification/badging process framework. This is a general framework to be followed with O-RAN certification/badging.



Figure 5.1-2: Certification/badging process framework –parts to be performed by OTICs

5.1.4 O-RAN Certificates and Badges

O-RAN certificates or badges issued by OTICs are comprised of:

- **Logo of the O-RAN Certificate, IOT Badge or E2E Badge**, which the product that is awarded the certificate or badge can display on the product or in corresponding marketing material, e.g. website, brochure, etc.
- **Summary report (O-RAN Certificate and Badge Summary Report)** providing a summary of O-RAN certification, IOT badging or E2E badging testing. The unified structure and format of document (using the same file format as associated template O-RAN.TIFG.Cert-Badge.1) is used to ensure reusability and sharing of the information. If CB-ICS Proforma is specified for a particular O-RAN

certificate, IOT badge or E2E badge, the Certification and Badging ICS shall be attached to the summary report.

- **Test report (O-RAN Certificate and Badge Test Report)** providing details of O-RAN certification, IOT badging or E2E badging testing. The unified format of document (associated template O-RAN.TIFG.Cert-Badge.2) is used to ensure reusability and sharing of the information
- Detailed **measurement results** providing a collection of detailed measurement results collected during the certification or badging including logs, screenshots, packet captures, etc. which are captured from DUT/SUT or/and from test and measurement equipment and tools. The file format of measurement results is not specified and unified and depends on used DUT/SUT and test and measurement equipment and tools.

The summary and test reports shall be encrypted, and editing shall be restricted to avoid unauthorized editing and modification of the reports. The summary and test reports shall be digitally signed by OTIC to provide assurance about integrity, validity, and authenticity of the reported data. It is recommended that the reports be co-signed by participating entity(ies) to confirm their involvement and their approval of reported data. The public key certificate (identity certificate) associated with the digital signature shall be valid and issued by a trusted certificate authority.

5.2 Processes

5.2.1 O-RAN Certification and Badging process flow

O-RAN Certification and Badging (C&B) process flow is illustrated in Figure 5.2.1-1.

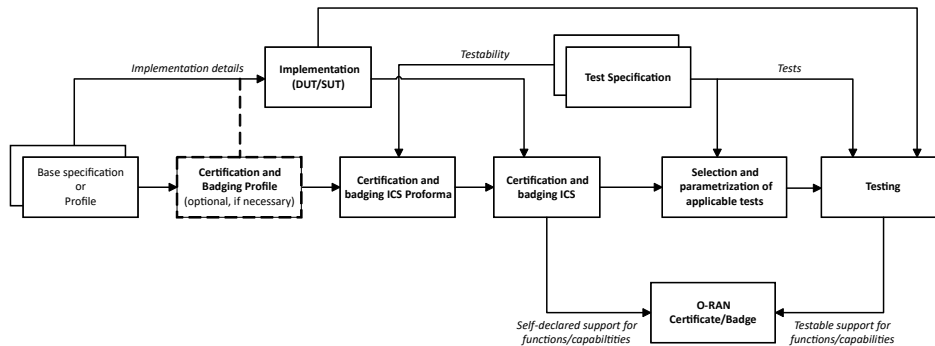


Figure 5.2.1-1: O-RAN Certification and Badging process flow diagram

For the purpose of O-RAN Certification and Badging and to facilitate O-RAN productization, the **O-RAN Certification and Badging Profile** (CB-Profile) may be used to constrain the existing options in base specification(s)/profile, but shall not impose requirements that conflict with the existing requirements in base specification(s)/profile (e.g. a function/capability/parameter mandatory in the base specification shall not be optional in the profile; or adding new function/capability/parameter that does not exist in the base specification is not allowed).

NOTE: The need for CB-profiles is expected to be relatively rare and will only be used to support the awarding of meaningful O-RAN certificates and badges.

1 The **Certification and Badging ICS Proforma (CB-ICS Proforma)** provides overview of all mandatory/optional/
2 conditional functions and capabilities defined in base specification(s)/profile and, if exists, in O-RAN
3 Certification and Badging Profile. It helps vendors ensure that their products meet the necessary
4 requirements for O-RAN Certificates and Badges. It helps to unambiguously and quickly identify the
5 mandatory/optional/conditional functions and capabilities. It also includes references to base specification(s)
6 where the implementation details are specified.

7 All functions and capabilities claimed as supported in the CB-ICS shall be tested if there is an applicable test.
8 Some functions and capabilities are self-declared for their support if they are not testable or tests for them
9 do not exist (e.g. version of IP protocol). Whether support for a particular function or capability shall be
10 verified by test(s) or can be self-declared is also specified in CB-ICS Proforma.

11 Vendors make an implementation (DUT/SUT) based on the base specification(s)/profile(s) and, if exists, on
12 the CB-profile. When the implementation is done, vendors use CB-ICS Proforma to declare which functions
13 and capabilities have been implemented/supported. When filled in, CB-ICS Proforma becomes Certification
14 and Badging ICS (CB-ICS).

15 The **Certification and Badging ICS (CB-ICS)** provides an overview of the implemented and supported
16 functions and capabilities for the purpose of O-RAN Certificates and Badges. It helps testers to unambiguously
17 select and parameterize the applicable tests based on functions and capabilities stated as implemented and
18 supported by the vendors. Tests are applicable for functions and capabilities specified as mandatory
19 (required) in base specification(s)/profile(s) and, if exists, in CB-profile, and for optional functions and
20 capabilities declared as implemented/supported by vendor(s). The CB-ICS should use a unified and machine-
21 readable format.

22 The Certification and Badging ICS Proforma in the form of questionnaire (checklist) should contain the
23 following data:

- 24 • unique identifier within the CB-ICS Proforma to identify the function/capability/parameter
- 25 • reference to clause(s) in base specification(s) describing the implementation details for each function
26 or capability
- 27 • whether the function or capability is mandatory/optional/conditional to be implemented/supported
28 for the purpose of O-RAN Certificate or Badge (Implementation requirements)
- 29 • whether the function or capability is supported/implemented or not
- 30 • whether implementation/support of function or capability shall be verified by test(s) (Testability).
31 The test(s) may not exist, or testing is not required.

32 The CB-Profiles (O-RAN.TIFG.TS.Cert-Badge.4) and CB-ICS Proformas (O-RAN.TIFG.TS.Cert-Badge.5) are
33 defined as standalone subsidiary documents to the present main document in the O-RAN TIFG in cooperation
34 with the relevant WG(s)/FG(s).

35 For O-RAN certificates, one CB-Profile and one CB-ICS Proforma are defined for each device and
36 interface/service under test, e.g. Open FH O-RU, Open FH O-DU. For O-RAN IOT Badge, one CB-Profile and
37 one CB-ICS Proforma are defined for each interface/service under test including both devices in the pair, e.g.
38 Open FH (including both O-RU and O-DU). For O-RAN E2E Badge, one CB-Profile and one CB-ICS Proforma are
39 defined for the whole system under test (not to individual devices and interfaces) and to E2E functionalities
40 and capabilities.

41 If CB-ICS Proforma is specified for a particular O-RAN certificate, IOT badge or E2E badge, the Certification
42 and Badging ICS (CB-ICS) is a part of O-RAN certificate, IOT badge or E2E badge, and shall be attached to the
43 summary report.

5.2.2 Defining Certification/Badging and Set of Tests

O-RAN Work Groups (WGs) specify Conformance Tests and Interoperability Tests related to their scope of work (e.g. WG4 specifies Conformance Tests and Interoperability Tests for Open Fronthaul interface). O-RAN TIFG specifies E2E tests where applicable.

O-RAN TIFG defines certification and badging processes and procedures while conformance, interoperability and E2E testing are defined in O-RAN WG/FG test specifications. This is specified in Chapter 6 for O-RAN certification, Chapter 7 for O-RAN IOT badging and Chapter 8 for O-RAN E2Ebadging.

O-RAN TIFG should define certification and badging according to market needs. If any OTIC does not find certificate/badge defined for its intended scenario(s), then the OTIC is encouraged to submit a proposal to the O-RAN TIFG in order to define new certificate/badge.

5.2.3 Test platforms

To reliably and successfully execute the tests for conformance certification, IOT badging, and E2E badging, specialized test equipment and tools are needed. Such test equipment and tools need to meet specific criteria and be designed to execute the identified tests.

It is the responsibility of OTICs to use appropriate test equipment and tools for certification/badging. O-RAN does not provide qualification or acknowledge processes for test equipment and tools, i.e. the choice of test equipment and tools for certification/badging will be left to the test market and their suitability will depend on the self-declaration of OTICs and test equipment vendors.

The OTICs should use commercially available or open-source test platforms. The OTICs should also ensure proper calibration of test equipment and tools as specified by the test equipment vendors. (Pre-) Commercial UE(s) can also be used as long as it is allowed in the associated O-RAN test specification(s) and the information about the UE(s) is documented as a part of the test setup and test results are reproducible.

The description of certification/badging test setup including information about used test equipment and tools needs to be documented in the summary report and test report. This will allow the requesting entity to challenge any failed test that may be caused by the test equipment.

5.2.4 Identifying Applicable O-RAN certificate/badge and Tests

The following steps describe the process for OTICs to identify which O-RAN certification/badging and test sets to conduct in the course of certification/badging of the DUTs/SUT. The process will take into consideration the interfaces and functions supported by the DUTs/SUT that are brought in for certification/badging.

Step 1: Identify which type of O-RAN certification/badging can be applied: Conformance certification (Chapter 6), Interoperability badging (Chapter 7) and/or E2E system integration badging (Chapter 8).

Step 2: Identify relevant DUT(s)/SUT and interface(s) to be tested based on O-RAN logical architecture [3] and type of O-RAN certification/badging identified in Step 1.

Step 3: Identify relevant test sets for certification or badging (see Chapters 6-8) of the specific DUT(s)/SUT and interface(s) identified in Step 2.

5.2.5 Executing Tests

The OTICs execute the identified test sets based on type of O-RAN certificate/badge and relevant DUT(s)/SUT identified in Chapter 5.2.3.

5.2.6 Verifying Test Results

The OTICs shall collect the test results for each executed test and fill in the summary report and test report to be shared accordingly. The detailed measurement results such as logs, screenshots, packet captures, etc. shall be also recorded. Each test in the summary report and test report must be evaluated against the expected results in associated test specification(s) to determine whether test passed or failed. The test results should be reproducible, i.e. the same results should be observed under the same conditions.

5.2.7 Issuing Certificates/Badges

The O-RAN certificate or badge can be awarded only to DUT (product) representing O-RAN function(s) (node(s)) (such as Near-RT RIC, O-CU-CP, O-CU-UP, O-DU, O-RU, etc.) with O-RAN interface(s) (such as Open Fronthaul, O1, A1, etc.) as defined in O-RAN architecture [3], or to 3GPP function(s) (node(s)) (such as eNB, gNB-CU, gNB-DU, etc.) with O-RAN profiled 3GPP interfaces (such as Xn, F1, etc.). The OTIC issues O-RAN certificate or badge to each individual DUT involved in the certification or badging. In case of O-RAN IOT badge, both DUTs involved in DUT pair are awarded by O-RAN IOT badge with the same certificate/badge reference identifier (see 5.2.6.1). In case of O-RAN E2E badge, all DUTs involved in E2E system (SUT) are awarded by O-RAN E2E badges with the same certificate/badge reference identifier. Assigning the same certificate/badge reference identifier helps to identify which DUTs have been involved in DUT pair for interoperability testing or E2E system for end-to-end testing.

5.2.7.1 Certificate/Badge Reference Identifier

The OTIC manages and assigns its own certificate/badge reference identifiers with the following naming convention. The naming convention is common for all O-RAN certificates, O-RAN IOT badges, and O-RAN E2E badges. The certificate/badge reference identifier is unique for each certification or badging testing. The certificate/badge reference identifier is a nine or ten-digit alphanumeric name with the following convention:

NAMEyy####

where:

NAME is unique 3 or 4 letter OTIC designator

yy are the last two digits of the year of issue

is sequential four-digit serial number within the year of issue (start from '0001') assigned individually by each OTIC

5.2.8 Compliance Folder

The Compliance Folder represents an electronic collection of all relevant documents for a product submitted for certification/badging. It is the responsibility of the requesting entity to create and maintain such a Compliance Folder, which must be kept for a period of no less than six years and must be available for sharing under the definition of described in Section 5.2.8. The Compliance Folder contents shall contain the certification and badging report template. It is recommended that requesting entity collect and store all relevant information in such a Compliance Folder.

The requesting entity appoints a person responsible within the requesting entity's organization for all matters related to the submission, execution, and maintenance of the product's certification. requesting entity shall provide evidence that a Quality Management System (QMS), i.e., ISO9001 or equivalent recognized standard, is followed. The compliance folder shall be uniquely identified, included in the requesting entity's document control system.

In the case of IOT or End-to-end badging process, where multiple requesting entities would participate in the tests, it is recommended each requesting entity create and maintain its own Compliance Folder independently with information relevant to its own products, subsystems or interfaces involved in such tests. Even though it is possible that some information stored is redundant, the independently maintained Compliance Folder by each requesting entity involved in the tests is necessary in case there is need to resolve data or result conflict or dispute. To further clarify, in the case that the requesting entity is the product owner but not the original equipment vendor/manufacture, it is the responsibility of the requesting entity for maintaining the Compliance Folder for all products involved.

5.2.9 Storing and Sharing of Summary Reports, Test reports and Measurement Results

The O-RAN public website keeps a catalogue with basic (non-sensitive) information (such as Certificate/Badge ID, Date of issued certificate/badge, Vendor name, OTIC name) about all issued O-RAN certificates and badges from all OTICs. This centralized catalogue has unrestricted access and is available to all O-RAN as well as non-O-RAN entities. Each individual OTIC shall promptly report to O-RAN Office the basic information needed for the O-RAN catalogue for any newly issued O-RAN certificate or badge. The information reported by OTICs is subject to the review and approval by O-RAN Office before publication.

The summary reports and test reports associated with the O-RAN certificate or badge shall be securely stored by the individual OTICs issuing the O-RAN certificate or badge. The detailed measurement results shall be stored either by the requesting entity (in case the measurement results are captured by the requesting entity from its DUT(s)) or by OTIC (in case the measurement results are captured by OTIC from its test and measurement equipment and tools) participating in the certification or badging testing.

Optionally, Annex D defines a machine-readable results artifact, that may be used by OTICs and others for the storage and handling of test results. This results artifact may be directly generated by automation tools

employed by the OTIC or by the test platforms (commercial or open source). Similarly, the OTIC may employ automation tools to produce the summary report or the test report from the results artifact.

Since O-RAN certificates and badges can only be issued by OTIC, the test results sharing principles defined for OTIC [1] need to be followed. On request, the OTIC shall share the summary reports and test reports (associated with O-RAN certificates and badges issued by the OTIC) only with O-RAN members (i.e. mobile network operators) and O-RAN Office. The OTIC shall not share the summary reports and test reports (associated with O-RAN certificates and badges issued by the OTIC) with other O-RAN entities (i.e. O-RAN contributors and academic contributors) as well as non-ORAN entities without the written permission of all copyright holders (i.e. the OTIC and all requesting entities participating in the certification or badging testing). The detailed measurement results may be shared at the discretion of either the requesting entity (the measurement results stored by the requesting entity) or OTIC (the measurement results stored by OTIC).

All sharing requests for the summary reports, test reports and measurement results should be directed to the OTIC that issued the O-RAN certificate or badge. In case of sharing request for measurement results, the OTIC forwards the request to the original participating entity(ies).

5.2.10 SH/HW changes to certified/badged devices

When SW/HW changes are made to a DUT/SUT, certification/badging for the previous versions of that DUT/SUT may not necessarily hold. On the other hand, a SW/HW change may also have nothing to do with interfaces which were the subject of previous certification/badging. The requesting entity should judge and self-declare when a redo of certification/badging is necessary with SW/HW changes.

5.2.11 Multi-vendor Scenario

The products under the O-RAN certification and badging process could present the following two multi-vendor scenarios:

- multi-vendor DUT: a single product with multiple internal SW/HW components (e.g. OS, HW, network cards, accelerator cards) from various vendors
- multi-vendor SUT: a combination of multiple DUTs (products) from same vendor or different vendors, where individual DUT (product) can be also multi-vendor DUT

In general, regardless of single vendor or multi-vendor DUT or SUT described above, O-RAN certificate or badge can be awarded only to DUT (product) representing O-RAN function(s) (node(s)) with O-RAN interface(s) or 3GPP function(s) (node(s)) with O-RAN profiled 3GPP interface(s) (see 5.2.6).

For the awarding of O-RAN certificates and badges in multi-vendor scenario, the following guidelines shall be observed:

- In the case of multi-vendor DUT described above, the single DUT (product) may be awarded the O-RAN certificate or badge although its internal SW/HW components could come from various vendors.
- In the case of multi-vendor SUT described above, all single or multi-vendor DUTs included in the end-to-end system (SUT) or interoperability DUT pair may be awarded the O-RAN badges.

Underlying OTICs and involved requesting entity(ies) shall record the information about all involved products and products' components in the final summary and test report defined by O-RAN certification and badging process.

5.3 Defined and Available O-RAN Certificates and Badges

Based on the current availability of Conformance/Interoperability/End-to-end test specifications from all O-RAN WGs/FGs, the O-RAN certificates and badges have been defined for the following combinations of Interface Under Test and Device Under Test and are ready for issuance under this release. More O-RAN certificates and badges will be defined for issuance in future releases when the corresponding test specifications become available. Note that the "RAT (Radio Access Technology) under test" is a RAT (5G NR or/and 4G LTE) which is supported by Device under test, and which has been also applied within the testing.

• Certificates (Conformance)

Interface / Service Under Test	Device Under Test	RAT Under Test	Notes
Open Fronthaul (OpenFH)	O-RU	4G LTE or/and 5G NR	
Open Fronthaul (OpenFH)	O-DU	4G LTE or/and 5G NR	
Open Fronthaul (OpenFH)	O-DU/O-CU	4G LTE or/and 5G NR	Combined O-DU and O-CU [3]
A1 services	Non-RT RIC	Not Applicable	
A1 services	Near-RT RIC	Not Applicable	

12

• IOT Badges

Interface / Service Under Test	Device Under Test	RAT Under Test	Notes
Open Fronthaul (OpenFH)	O-RU	4G LTE or/and 5G NR	
Open Fronthaul (OpenFH)	O-DU	4G LTE or/and 5G NR	
Open Fronthaul (OpenFH)	O-DU/O-CU	4G LTE or/and 5G NR	Combined O-DU and O-CU [3]
X2	eNB	4G LTE	3GPP compliant node
X2	en-gNB	5G NR	3GPP compliant node [6]
F1	gNB-CU	5G NR	3GPP compliant node [5]
F1	gNB-DU	5G NR	3GPP compliant node [5]
Xn	gNB	5G NR	3GPP compliant node [4]
A1 services	Non-RT RIC	Not Applicable	
A1 services	Near-RT RIC	Not Applicable	

14

• E2E Badges

Interface Under Test	Device Under Test	RAT Under Test	Notes
Not Applicable	O-RU	4G LTE or/and 5G NR	
Not Applicable	O-DU	4G LTE or/and 5G NR	
Not Applicable	O-CU	4G LTE or/and 5G NR	
Not Applicable	O-RU/O-DU	4G LTE or/and 5G NR	Combined O-RU and O-DU [3]
Not Applicable	O-DU/O-CU	4G LTE or/and 5G NR	Combined O-DU and O-CU [3]
Not Applicable	O-RU/O-DU/O-CU	4G LTE or/and 5G NR	Combined O-RU, O-DU and O-CU [3]

15

6. O-RAN Certificate Procedures

This section provides the procedures for verification of compliance of O-RAN interface implementations with O-RAN defined interface specifications resulting in the award of O-RAN certificate.

6.1 Subsystems and Interfaces

The O-RAN conformance test specifications have been defined for the following combinations of O-RAN specified interface and subsystem (device under test), and these combinations can be awarded an O-RAN certificate. Table 6.1-1 will be updated following publication of new O-RAN conformance test specifications, and as soon as new combinations of interfaces and subsystems can be awarded an O-RAN certificate.

Table 6.1-1: The combinations of interface and subsystem (device under test) which can be awarded by O-RAN certificate, incl. the applicable test specifications

Interface / Service	Subsystem (DUT)	Applicable test specification(s) / CB-Profiles / CB-ICS Proforma	O-RAN WG/FG owning test specification
O-RAN Open Fronthaul incl. CUSM-planes	O-RU	Open Fronthaul conformance test specification (O-RAN.WG4.CONF)	O-RAN WG4
O-RAN Open Fronthaul incl. CUSM-planes	O-DU	Open Fronthaul conformance test specification (O-RAN.WG4.CONF)	O-RAN WG4
O-RAN Open Fronthaul incl. CUSM-planes	O-DU/O-CU	Open Fronthaul conformance test specification (O-RAN.WG4.CONF)	O-RAN WG4
A1 services	Non-RT RIC	A1 interface Test Specification (O-RAN.WG2.TS.A1TS) O-RAN CB Profiles (O-RAN.TIFG.TS.Cert-Badge.4) O-RAN CB-ICS Proformas (O-RAN.TIFG.TS.Cert-Badge.5)	O-RAN WG2 O-RAN TIFG O-RAN TIFG
A1 services	Near-RT RIC	A1 interface Test Specification (O-RAN.WG2.TS.A1TS) O-RAN CB Profiles (O-RAN.TIFG.TS.Cert-Badge.4) O-RAN CB-ICS Proformas (O-RAN.TIFG.TS.Cert-Badge.5)	O-RAN WG2 O-RAN TIFG O-RAN TIFG

If needed, the following sections provide details on each combination mentioned in Table 6.1-1.

6.2 O-RAN Open Fronthaul interface of O-RU

All tests for validation of conformance of O-RAN specified Open Fronthaul interface of O-RU against relevant O-RAN technical specification are specified in the O-RAN "Open Fronthaul conformance test specification" (O-RAN.WG4.CONF). Tests that are mandatory are for those functionalities of the O-RU that by specification are required to be supported. Optional functionality in an O-RU implies that the corresponding tests are conditionally mandatory and are mandatory only if the O-RU supports the optional functionality. **All M-plane**

and S-plane tests marked as mandatory (M), and all applicable M-plane and S-plane tests marked as conditional mandatory (CM) shall be passed for issuing of the O-RAN certificate. For the selected test scenario, all UC-plane tests marked as mandatory, and all applicable UC-plane tests marked as conditional mandatory shall be passed for issuing of the O-RAN certificate. The user/tester identifies all conditionally mandatory tests based on the optional functionalities that the O-RU declares as supported. The user/tester also selects one applicable test scenario that can be executed for UC-plane tests. The tests marked as optional (O) or other unmarked tests from the test specification or any tests from other sources (e.g. 3GPP specifications) can be executed, and test results may be added to summary report and test report, but passing of these tests is not required for issuing of the specific O-RAN certificate.

For the purpose of O-RAN Certificate, all the information and data from the testing shall be collected in the summary report (tab "Certificate O-RU OpenFH") and test report.

6.3 O-RAN Open Fronthaul interface of O-DU

All tests for validation of conformance of O-RAN specified Open Fronthaul interface of stand-alone O-DU (or combined O-DU/O-CU) against relevant O-RAN technical specification are specified in the O-RAN "Open Fronthaul conformance test specification" (O-RAN.WG4.CONF). Tests that are mandatory are for those functionalities of the O-DU that by specification are required to be supported. Optional functionality in an O-DU implies that the corresponding tests are conditionally mandatory and are mandatory only if the O-DU supports the optional functionality. **All tests marked as mandatory (M) and all applicable tests marked as conditional mandatory (CM) shall be passed for issuing of O-RAN certificate.** The user/tester identifies all conditionally mandatory tests based on the optional functionalities that the O-DU declares as supported. The tests marked as optional (O) or other unmarked tests from the test specification or any tests from other sources (e.g. 3GPP specifications) can be executed, and test results may be added to summary report and test report, but passing of these tests is not required for issuing of the specific O-RAN certificate.

For the purpose of O-RAN certificate, all the information and data from the testing shall be collected in the summary report (tab "Certificate O-DU OpenFH") and test report.

6.4 O-RAN A1 services on A1 interface of Non-RT RIC

The certificate on A1 interface of Non-RT RIC is currently defined at the service level only, and does not include the entire A1 interface, which contains additional functionalities such as functions at transport layer (A1TP). A1 services on A1 interface of Non-RT RIC refer to A1 policy management service (A1-P) and A1 enrichment information service (A1-EI).

All tests for validation of conformance of O-RAN A1 services on A1 interface of Non-RT RIC against relevant O-RAN technical specifications are specified in O-RAN.WG2.TS.A1TS. O-RAN CB-Profile (O-RAN.TIFG.TS.Cert-Badge.4) further constrains the options specified in the base specification. The overview of all mandatory/optional/conditional functions and capabilities is provided in O-RAN Certification and Badging ICS (CB-ICS) Proforma (O-RAN.TIFG.TS.Cert-Badge.5).

O-RAN certificate of A1 services on A1 interface of Non-RT RIC is issued as a single certificate without the underlying supported A1 service(s) explicitly specified. The relevant information on specific A1 service(s) and procedures supported by Non-RT RIC is recorded and described in the associated Summary Report and Test Report.

Commented [JP1]: The name of clause is "O-RAN A1 services ..."

6.5 O-RAN A1 services on A1 interface of Near-RT RIC

The certificate on A1 interface of Near-RT RIC is currently defined at the service level only, and does not include the entire A1 interface, which contains additional functionalities such as functions at transport layer (A1TP). A1 services on A1 interface of Near-RT RIC refer to A1 policy management service (A1-P) and A1 enrichment information service (A1-EI).

All tests for validation of conformance of O-RAN A1 services on A1 interface of Near-RT RIC against relevant O-RAN technical specifications are specified in O-RAN.WG2.TS.A1TS. O-RAN CB-Profile (O-RAN.TIFG.TS.Cert-Badge.4) further constrains the options specified in the base specification. The overview of all mandatory/optional/conditional functions and capabilities is provided in O-RAN Certification and Badging ICS (CB-ICS) Proforma (O-RAN.TIFG.TS.Cert-Badge.5).

O-RAN certificate of A1 services on A1 interface of Near-RT RIC is issued as a single certificate without the underlying supported A1 service(s) explicitly specified. The relevant information on specific A1 service(s) and procedures supported by Near-RT RIC is recorded and described in the associated Summary Report and Test Report.

7. O-RAN IOT Badge Procedures

This section provides the procedures for validation of interoperability (IOT) of subsystems utilizing O-RAN specified or profiled open interfaces resulting in the award of O-RAN IOT badge.

7.1 Subsystems and Interfaces

The O-RAN interoperability test specifications have been defined for the following combinations of O-RAN specified interface and subsystems (device under test), and these combinations can be awarded an O-RAN IOT Badge. The Table 7.1-1 will be updated following publication of new O-RAN interoperability test specifications, and as soon as new combinations of interfaces and subsystems can be awarded an O-RAN IOT Badge.

Table 7.1-1 The combinations of interface and subsystems (devices under tests) which can be awarded by O-RAN IOT Badge, incl. the applicable test specifications

Interface / Service Under Test	Subsystems (DUT pair)	Applicable test specification(s) / CB-Profiles / CB-ICS Proformas	O-RAN WG/FG owning test specification
O-RAN Open Fronthaul	O-RU, O-DU	Fronthaul Interoperability Test Specification (IOT) (O-RAN.WG4.IOT)	O-RAN WG4
O-RAN Open Fronthaul	O-RU, O-DU/O-CU	Fronthaul Interoperability Test Specification (IOT) (O-RAN.WG4.IOT)	O-RAN WG4
X2	eNB, en-gNB	Interoperability Test Specification (IOT) (O-RAN.WG5.IOT)	O-RAN WG5
F1	gNB-CU, gNB-DU	Interoperability Test Specification (IOT) (O-RAN.WG5.IOT)	O-RAN WG5

Xn	gNB, gNB	Interoperability Test Specification (IOT) (O-RAN.WG5.IOT)	O-RAN WG5
A1 services	Non-RT RIC. Near-RT RIC	A1 interface Test Specification (O-RAN.WG2.TS.A1TS) O-RAN CB Profiles (O-RAN.TIFG.TS.Cert-Badge.4) O-RAN CB-ICS Proformas (O-RAN.TIFG.TS.Cert-Badge.5)	O-RAN WG2 O-RAN TIFG O-RAN TIFG

1 If needed, the following sections provide details on each combination mentioned in Table 7.1-1.

2 7.2 O-RAN Open Fronthaul interface

3 All tests for validation of interoperability between O-RU and stand-alone O-DU (or combined O-DU/O-CU)
4 utilizing O-RAN Open fronthaul interface are specified in the O-RAN “Fronthaul Interoperability Test
5 Specification (IOT)” (O-RAN.WG4.IOT). **All tests marked as mandatory (M) and all applicable tests marked
6 as conditional mandatory (CM) shall be passed for issuing of O-RAN IOT badge.** The optional tests or other
7 unmarked tests from the test specification or any tests from other sources (e.g. 3GPP specifications) can be
8 executed, and test results may be added to summary report and test report, but passing of these tests is not
9 required for issuing of the specific O-RAN IOT badge.

10 The Fronthaul Interoperability Test Specification (IOT) (O-RAN.WG4.IOT) defines Interoperability Test (IOT)
11 Profiles along with associated IOT Profile Test Configurations. For O-RAN IOT badge, both O-RU and O-DU (or
12 combined O-DU/O-CU) need to agree and use at least one common IOT Profile Test Configuration or
13 Customized IOT Configuration to verify interoperability. Further, for O-RAN IOT Badge, the participating
14 entity can choose one or both of the following options:

- 15 • Option 1: selection of **IOT Profile Test Configuration(s)** listed in the Fronthaul Interoperability Test
16 Specification (IOT) (O-RAN.WG4.IOT). The name(s) of used IOT Profile Test Configuration shall be
17 mentioned in the summary report (tab “IOT Badge OpenFH”).
- 18 • Option 2: creation of **Customized IOT Configuration(s)**. The Customized IOT Configuration(s) (incl.
19 the values of all required system parameters) shall be included in the summary report (tab “OpenFH
20 Customized IOT Conf.”). In case of multiple Customized IOT Configurations, individual column shall
21 be created for each Customized IOT configuration. The name(s) of used Customized IOT
22 Configuration shall be mentioned in the summary report (tab “IOT Badge OpenFH”).

23 The IOT Profile Test Configuration(s) (PTC) may contain a list of “to be validated” capabilities. The utilization
24 of “to be validated” capabilities associated with each PTC shall be validated in C/U-plane IOT tests for issuing
25 of O-RAN IOT badge. The IOT PTC may also contain a list of “to be reported” capabilities. The usage, or
26 absence of usage, of these capabilities shall be reported in the test report for issuing of O-RAN IOT badge.

27 For O-RAN IOT Badge, all the information and data from the testing shall be collected in the summary report
28 (tab “IOT Badge OpenFH”) and test report.

29 7.3 X2 interface

30 All tests for validation of interoperability between subsystems (DUT pairs listed in Table 7.1-1) utilizing X2
31 interface are specified in the O-RAN “Interoperability Test Specification (IOT)” (O-RAN.WG5.IOT). Based on
32 the supported capabilities, functionalities, and features of Devices under Test (DUTs), the user/tester needs

to identify all applicable X2 C-plane and X2 U-plane IOT tests from this test specification. **All applicable X2 C-plane and U-plane IOT tests shall be passed for issuing of O RAN IOT badge.**

For the purpose of O-RAN IOT Badge, all the information and data from the testing shall be collected in the summary report (tab "IOT Badge X2") and test report.

7.4 F1 interface

All tests for validation of interoperability between subsystems (DUT pairs listed in Table 7.1-1) utilizing F1 interface are specified in the O-RAN "Interoperability Test Specification (IOT)" (O-RAN.WG5.IOT). Based on the supported capabilities, functionalities, and features of Devices under Test (DUTs), the user/tester needs to identify all applicable F1 IOT tests from this test specification. **All applicable F1 IOT tests shall be passed for issuing of O-RAN IOT badge.**

For the purpose of O-RAN IOT Badge, all the information and data from the testing shall be collected in the summary report (tab "IOT Badge F1") and test report.

7.5 Xn interface

All tests for validation of interoperability between subsystems (DUT pairs listed in Table 7.1-1) utilizing Xn interface are specified in the O-RAN "Interoperability Test Specification (IOT)" (O-RAN.WG5.IOT). Based on the supported capabilities, functionalities, and features of Devices under Test (DUTs), the user/tester needs to identify all applicable Xn IOT tests from this test specification. **All applicable Xn IOT tests shall be passed for issuing of O-RAN IOT badge.**

For the purpose of O-RAN IOT Badge, all the information and data from the testing shall be collected in the summary report (tab "IOT Badge Xn") and test report.

7.6 A1 services on A1 interface

The IOT badge on A1 interface is currently defined at the service level only between Non-RT RIC and Near-RT RIC (DUT pairs listed in Table 7.1-1), and does not include the entire A1 interface, which contains additional functionalities such as functions at transport layer (A1TP). A1 services on A1 interface refer to A1 policy management service (A1-P) and A1 enrichment information service (A1-EI).

All tests for validation of interoperability between Non-RT RIC and Near-RT RIC utilizing A1 services are specified in O-RAN.WG2.TS.A1TS. O-RAN CB-Profile (O-RAN.TIFG.TS.Cert-Badge.4) further constrains the options specified in the base specification. The overview of all mandatory/optional/ conditional functions and capabilities is provided in O-RAN Certification and Badging ICS (CB-ICS) Proforma (O-RAN.TIFG.TS.Cert-Badge.5).

O-RAN IOT badge for A1 services on A1 interface between Non-RT RIC and Near-RT RIC is issued as a single badge without the underlying supported A1 service(s) explicitly specified. The relevant information on specific A1 service(s) and procedures supported by Non-RT RIC and Near-RT RIC is recorded and described in the associated Summary Report and Test Report.

8. O-RAN E2E Badge Procedures

This section provides the procedures for validation of end-to-end functionality and security of end-to-end systems or subsystems utilizing O-RAN specified or profiled open interfaces resulting in the award of O-RAN E2E badge.

All tests for validation of end-to-end functionality are specified in O-RAN End-to-end Test Specification (O-RAN.TIFG.E2E-Test) and for end-to-end security in O-RAN Security Test Specifications (O-RAN.WG11.SecTestSpecs). **All applicable functional tests and “3GPP Security Assurance Specification (SCAS)” tests shall be passed for issuing of O-RAN E2E badge.** The applicability of tests depends on the supported capabilities, functionalities, and features of Device(s) under test/System under test. The other tests from the test specification or any tests from other sources (e.g. 3GPP specifications) can be executed, and test results may be added to summary report and test report, but passing of these tests is not required for issuing of the specific O-RAN E2E badge.

For the purpose of O-RAN E2E badge, the participating entity can choose one of the following scenarios:

- **Validation of E2E functionality and security of single or multiple Device(s) under test (DUTs) as a part of E2E reference test setup (subsystem replacement testing)**
- **Validation of E2E functionality and security of the entire E2E test setup as System under test (SUT)**

For the purpose of O-RAN E2E Badge, all the information and data from the testing shall be collected in the summary report (tab “E2E Badge”) and test report.

Annex A (informative):

Certification application

Certification application should at least include the following information:

- Company
- Contact Name
- Contact e-mail
- Contact Address
- Contact Phone
- Product Name/Model
- Product Software/Firmware Version
- Product Description
- Date of Submission
- Date of expected start of testing
- Preferred OTIC
- Sub-system Type

If the product uses certified building blocks or wishes to reuse results from an already certified/tested product, provide a description of the sub-assembly/building block and identify the source certified/tested product(s). Include any potential differences, changes or other factors that may impact the definition of needed regression testing.

Annex B (informative):

Compliance folder

Compliance folder should at least include the following information:

- Index of contents
- Completed ICS with test execution information
- Model/type numbers
- Brand name
- Manufacturer detail
- Technical description including a block diagram, circuit diagram
- Details of significant design aspects (features, specifications, installation measures, control of variants and manufacturing)
- Variant description (as applicable)
- Software version(s) / SVN
- Reference to versions of O-RAN ALLIANCE Certification Criteria used
- User, installation, or special instructions
- Test data
- Internal and the third party test reports
- Products tested for interoperability
- Details of changes in hardware and/or software configuration together with evidence of continued compliance
- Evidence that the device has been developed and tested according to the Quality Management Program requirements
- A copy of the Certification Declaration

Annex C (normative): Trademark License Agreement

Trademark License Agreement

between

O-RAN ALLIANCE e.V., Buschkauler Weg 27, 53347 Alfter, Germany

- hereinafter referred to as "Licensor" -

and

<entity without O-RAN membership (legal form, address)>

- hereinafter referred to as the "Licensee" –

Preamble

The background of the Trademark License Agreement is to use O-RAN ALLIANCE trademarks for O-RAN Certificates and Badges by non O-RAN ALLIANCE members. The aim of the Trademark License Agreement is to allow the use of the O-RAN ALLIANCE trademarks **"O-RAN Certified", "O-RAN Interoperable" and "O-RAN E2E-Tested"** with certified and/or badged products.

§ 1 License

(1) The Licensor has been granted by the owner of the trademarks "O-RAN Certified", "O-RAN Interoperable" and "O-RAN E2E-Tested" and registered in the trademark register at the [...] on [...] under No. [...] for [will be detailed by O-RAN] (hereinafter "the contractual trademark").

(2) The Licensor hereby grants the Licensee the non-exclusive and non-transferable right, for the duration of the Agreement, in the territory of [...] to offer, place on the market, advertise, use on business papers and to label these products (hereinafter referred to as "the Contract Products") with the contractual trademark [...].

(3) The Licensee is not entitled to grant sub-licenses.

§ 2 License endorsement

(1) The Licensee undertakes to use the contractual trademark, in particular on labels, packaging and in advertising, only in the form prescribed by the Licensor and, if the Licensor deems additions necessary, only with the additions prescribed by the Licensor. The Licensee shall always mark the contractual trademark with the registration mark ® and indicate that it is a registered trademark of the Licensor.

(2) The Licensor shall notify the Licensee in writing of any changes to the prescribed form or additions and these shall be implemented without delay, but at the latest within the period set by the Licensor. After expiry of the deadline set by the Licensor, the Licensee may only use the contractual trademark in a non-updated form or with non-updated additions with the written consent of the Licensor.

§ 3 Warranty and liability

(1) The Licensee shall indemnify the Licensor and hold the Licensor harmless from all third-party claims arising from and in connection with the contractual services, unless the Licensor is at least predominantly responsible for the claims and liabilities. The Licensor shall be predominantly responsible for damages which have come about as a result of its rights to issue instructions under this Agreement. Otherwise, Section 254 German Civil Code applies.

(2) The Licensor does not guarantee that the use of the contractual trademark does not infringe any third-party rights. However, the Licensor declares that he is not aware of any such rights. No liability is assumed for freedom from defects, in particular the dependence on third-party property rights.

(3) The Licensor warrants that the contractual trademark has not been pledged, that no rights of use or other rights in rem have been granted and that the contractual trademark is not the subject of enforcement proceedings and is not subject to any other ongoing proceedings.

§ 4 Contract duration and termination

(1) This contract shall enter into force on [...] and has a term until [...]. Thereafter, it shall be automatically extended for a further [...] years, unless it is renewed with a notice period of [...] months to the end of the respective contract period.

(2) The contract shall end without the need for cancellation when the contractual trademark is legally cancelled.

(3) Irrespective of the other ordinary termination provisions or termination events regulated in the contract, this contract may be terminated by the Licensor with immediate effect for the following reasons:

(a) if the Licensee, its commercial agents, wholesalers and importers use the contractual trademark for products/services that are not covered by this Agreement or to which the Licensor has not consented, unless such use is discontinued immediately, at the latest within [...] calendar days after receipt of a written request to cease and desist.

(b) if the Licensee breaches its material contractual obligations or any other material provision of this Agreement, unless such breach occurs within [...] days after written request by the Licensor.

(c) if the Licensee is liquidated or insolvency proceedings are opened against the Licensee's assets or the opening of insolvency proceedings is rejected for lack of assets or similarly serious proceedings are initiated against the Licensee.

§ 5 Settlement after termination of the contract

Upon termination of this Agreement, the Licensee shall immediately refrain from using the contractual trademark; the Licensor shall remain the sole owner of all rights to the contractual trademark in relation to the Licensee. If the Licensee has acquired any rights to the trademark due to the use of the trademark, the Licensee shall transfer these rights to the Licensor upon termination of the Agreement. The Licensor accepts this transfer.

§ 6 Maintenance and defence of the contractual trademark

The Licensor undertakes to maintain the registration of the contractual trademark for the duration of this agreement and to defend it against any attacks by third parties. This does not apply if the acts of infringement are insignificant and the litigation costs would be disproportionate to any loss of sales. The assessment of whether an act of infringement is insignificant is the sole responsibility of the Licensor.

§ 7 Attacks by third parties against the use of the trademark by the Licensee

(1) If one of the parties becomes aware that a third party is using a labelling and/or registers it as a trademark which could possibly be confused with the contractual trademark or otherwise infringes it, it must inform the other party of this immediately. If a claim is made against the Licensee for injunctive relief and/or damages due to the manufacture of the products and/or the use of the contractual trademark by a third party, the Licensee is obliged to inform the Licensor of this immediately, providing the necessary details about the nature of the claim.

(2) In principle, the Licensee is not entitled to bring an action in its own name for trademark infringement. The Licensor is primarily entitled to take action against the use of a confusingly or otherwise infringing name. The Licensor can make the action dependent on the Licensee contributing up to half of the costs of the action. The right of the Licensee to join an infringement action brought by the trademark proprietor pursuant to Section 30 (4) MarkenG in order to claim damages remains unaffected. If the Licensor does not take action against the third party within a reasonable period of time despite the Licensee's offer to contribute half of the costs of the action, the Licensor is obliged to authorise the Licensee to pursue the infringement in his own name and at his own expense. Compensation paid by a third party shall be due to the parties in the proportion in which they have borne the costs of the action.

(3) The Licensor requires the consent of the Licensee to conclude a settlement by which he authorises a third party to use a trademark which, in the opinion of the Licensee, is likely to be confused with the contractual trademark.

§ 8 Non-aggression clause

The Licensee undertakes neither to attack the contractual trademark itself nor to support third parties in attacking property rights.

§ 9 License fee

The license will be granted free of-charge without any royalties.

§ 10 General provisions

(1) Should a provision of this contract be or become invalid, this shall not affect the validity of the remaining provisions. The parties undertake to replace the invalid provision with another valid provision that comes closest to the intended economic purpose. The same applies in the event of a gap in the contract.

(2) All amendments and additions to this contract must be made in writing to be effective. This shall also apply to any amendment or cancellation of this written form clause.

(3) This contract shall be governed by German law. The exclusive place of jurisdiction is the Regional Court [...].

Licensor**Licensee**_____
Name_____
Job title_____
Name_____
Job title_____
Place_____
Date_____
Place_____
Date_____
Signature_____
Signature

Annex D (normative):

Machine readable results artifact

This annex describes the architecture and requirements for the machine readable Results Artifact that is created as the outcome of testing according to O-RAN Alliance test specification or similar specifications. Implementation of the Results Artifact is optional, however, if implemented, the Results Artifact shall conform to the requirements within this Annex. Figure D.1 describes the high-level view of the information created during testing, and thus “stored” as part of the Results Artifact.

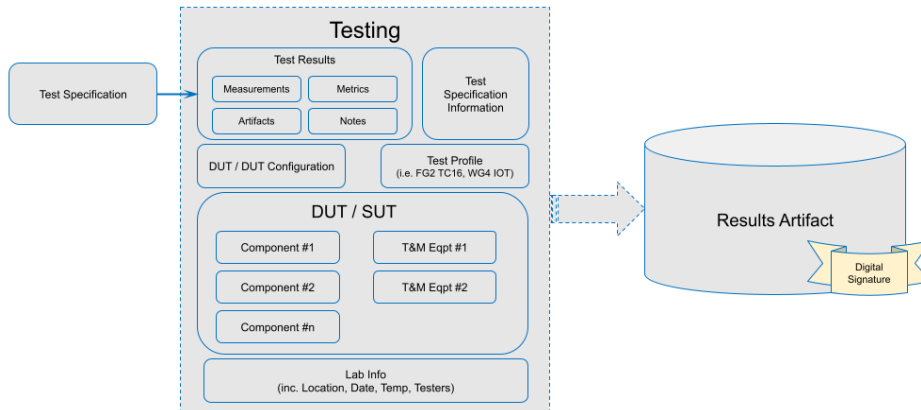


Figure D.1: Architecture and key components stored within the Results Artifact

D1 Definitions

- **Artifact Container** – A directory or compress archive (zip or tarball), containing a Results Artifact of a single run (one execution) of a Test Specification against one SUT or DUT.
- **Measurement** – Value or observable result, recorded during the execution of a test procedure.
- **Metadata (of test)** – General information about the test execution, such as start time, end time, overall outcome, identification numbers, etc.
- **Metric (Test Requirements)** – Measurable outcome of an item within a test case, resulting in a pass or fail determination. Note, a single Test Case can include multiple Test Requirements.
- **Results Artifact** – Outcome or deliverable of testing, including details of the system or device under testing, test and measurement equipment, laboratory, dates of testing, and measurement data. A Results Artifact should include enough information to ensure the testing could be replicated with the same conditions, hardware, and firmware/software versions, as a mechanism to ensure the traceability of the testing, and it can be used by a tool to automatically generate Summary and/or Test Reports.

- 1 • **Schema** – Language definition and model, encapsulating requirements, parameters, and syntax of
2 machine readable information within the Results Artifact.
- 3 • **T&M Equipment** – Equipment used within testing to provide test stimulus or measurements
4 according to the test plan requirements.
- 5 • **Test Case** – A construct within a Test Specification that includes at least a procedure and one or
6 more metrics, with the intention to yield an outcome (i.e. a pass or fail result). A Test Specification
7 typically includes one or more Test Cases, which might be organized into groups based on topic or to
8 promote overall readability of the document.
- 9 • **Testbed Component** – A unit or product, included within the network during testing, where the
10 behavior, performance, or operation of the unit or product could influence the outcome of testing.
11 Note, testing is likely performed on multiple interconnected components, where can all be included
12 in the Results Artifact.
- 13 • **Test Lab** – Organization responsible for the execution of the Test Specification, along with the
14 capture and documentation of the Test Results produced from that execution. Test Labs will
15 generally create the Results Artifact described within this annex.
- 16 • **Test Notes** – General information, observations, or other documentation developed and captured
17 during the execution of a Test Specification.
- 18 • **Test Specification** – A version-controlled document of test procedures, test requirements, and
19 required measurement or observable results that provides detailed guidance to laboratory
20 personnel responsible for the operation and execution of testing.
- 21 • **Test Results** – Collection output from the execution of a Test Specification, including Test
22 Measurements, Test Metrics, Artifacts, and other data collected during the testing process.

23 D2 Theory of Operation

24 The Results Artifact is made up of a collection of files, containing information about the testing conducted
25 by the test lab. A commonly named, results.json (or results.yaml) file, stored at the root of the file
26 connection serves as the machine readable entry point to the Results Artifact. The other files can be
27 organized into one or more sub-directories, but all files referenced within the Results Artifact are stored in
28 this location. This ensures there are no references to external materials needed for the interpretation or
29 review of the Test Results and its outcome(s). The machine readable results file contains, high level
30 metadata type information about testing, information about the device or systems under test, test and
31 measurement equipment, the lab organization responsible for the testing, test results and measurements,
32 as well as contact information for the parties involved the testing.

33 The schema also provides a means to refer to and describe the contents of other files stored within the
34 Results Artifact. These files are typically referred to as opaque files, where the format may or may not be a
35 commonly understood by the community. For example, a file might be a packet capture file (i.e. pcap
36 format), or it might be a binary file, specific to a test instrument, such as a spectrum analyzer. These
37 referenced files can also be utilized to store information about the configuration of the components of the
38 testbed. Lastly, the schema also provides a means to reference publicly available materials, such as test
39 specifications published by the O-RAN ALLIANCE or 3GPP.

40 It is anticipated the Results Artifact will be stored as a whole unit, either as a directory or compressed
41 archive. When the results are read by a machine, it is expected the machine will be provided with the path
42 to the Artifact Container.

D3 Requirements

D3.1 Artifact Container

1. The Results Artifact shall be represented at the root, as a single directory, that MUST contain one or more files or sub-directories.
2. There shall be at least one file in the root of the directory of the results artifact named either “results.yaml” or “results.json”. This file shall conform to the JSON schema <https://o-ran.org/schemas/o-ran-test-results>, as described below.
3. The Results Artifact shall contain the complete set of information from the testing, without references to external material, except for references to publicly available materials (i.e. Published O-RAN Alliance test plans) and captured as “decorated links” within the schema.

The contents of the Artifact Container can be used for the generation of the Summary Report and Test Reports in an automated manner, but the total contents of the Results Container are not limited to only the information required for the generation of those reports or formats.

D3.2 Test Results Schema

The JSON schema, identified as <https://o-ran.org/schemas/o-ran-test-results>, describes the complete information schema for the results file stored within the Test Artifact. This file will store information within a number of objects, each defined within the schema for a specific purpose. The sections below provide additional information and context about each object within the schema. Nothing within this document shall imply or cause an alteration to the interpretation of the actual JSON schema file, such as changing a value within the “required” parameter or any such similar change.

The results schema and sub-schemas are stored within the following files:

JSON Schema File Name	Description
o-ran-test-results-schema.json	Overall schema used to store information about the testing within the results artifact.
o-ran-wg4-iot-profile-schema.json	Values used within the IOT profile parameters, as defined by O-RAN ALLIANCE WG4.
o-ran-config-parameters-schema.json	Various configuration parameters that may be applied during testing.

D3.2.1 Schema Version

The version shall identify the version of the schema applied to the data, as a constant value.

D3.3.2 Test Metadata

The test metadata object contains high level information about the test, with mandatory objects: start date, device under test (DUT) name, overall result, test type, and the test ID.

1 The DUT name shall reference a uniquely identifiable component within the testbed components object of
2 the results. The DUT name may reference multiple components, if applicable to the type of test, such as the
3 case for interoperability testing.

4 The result shall indicate the aggregate result of the testing contained within the Results Artifact. The
5 overall, aggregated, test result. PASS indicates all required test cases also indicate PASS. FAIL indicates one
6 or more required test cases indicate FAIL. WARN indicates behavior observed during the execution of the
7 test case might cause concern, problems, or issues not directly relating to the required test metrics. SKIP
8 should not be used.

9 The test ID value shall be applied to all Results Archives created by the laboratory and shall be unique
10 across all Results Archives created by the laboratory.

11 D3.2.3 Testbed Components

12 The testbed components property contains an array of objects, where each object describes a device,
13 product, system, or test and measurement equipment used within the testing. One or more of these
14 components are the device or system under test, identified by the test metadata object.

15 Each component shall include the following properties: component description, manufacturer name,
16 manufacturer model. The object shall also include at least one of the following properties: software
17 version, firmware version, or hardware version.

18 The component description should clearly indicate the component role within the testing, if possible, with
19 reference the O-RAN architecture.

20 D3.2.4 Test Lab

21 The test lab object describes the laboratory responsible for the results contained within the Results Artifact.
22 The object shall provide at least one contact object, that describes a person, along with their details: email
23 and phone number. The object may provide one or more decorated links to public information about the
24 laboratory, such as the laboratory website.

25 D3.2.5 Test Specification

26 The test specification property contains an array of test specification objects, that shall uniquely identify at
27 least one test specification used to produce the Test Results contained within the Results Artifact.

28 D3.2.6 Tags

29 The tags property contains an array of unique strings, that may be utilized to describe the testing data
30 contained within the Results Artifact. Each string is anticipated as a short, free form string, that could be
31 used for results filtering, search, or similar operations within a database of results.

32 D3.2.7 Notes

33 The notes property contains an array of note objects. Each note shall contain a title and the text content of
34 the note.

1 D3.3.8 Test Results

2 The test results property contains an array of objects. Each object may be either a test group object or a
3 test case object. A test group object contains an array of test group or test case objects. Both the test group
4 and test case objects contain a required property to record the test number. In this way, the structure is
5 intended to mirror the structure and organization of the test specification within the test results.

6 The test case object includes required properties: number, name, description, result, status, and metrics.
7 Additionally, and number optional properties are defined, including contacts, start date, stop date, notes,
8 measurements, decorated links, and artifacts.

9 The metrics property contains an array of metrics objects, where each object described a test metrics, also
10 known as a test requirement, including the outcome result of the requirement and measurement values
11 required to validate that outcome.

12 The measurements property contains an array of measurement value objects. These measurements are in
13 addition to those used directly within the metrics objects.

14 D4 Digital Signature

15 For further study.

16

17

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- 1 Annex (informative):
 - 2 Change history/Change request (history)

Date	Revision	Description
2021.03.05	01.00	First published version
2021.07.09	02.00	1. Clarification of certification/badging objectives in Chapter 4 & test result sharing in Chapter 5 2. Alignment with the updated Conformance and IOT specs from WG4 and WG5 in Chapter 6 & 7 3. Inclusion of custom IOT profile in the badging process in Chapter 7 4. Making security test mandatory in E2E badging process in Chapter 8 5. Alignment with the updated E2E Test Spec by adding load & stress test and additional service tests
2021.11.10	03.00	1. Clarification in Chapter 4 on what entity can issue certificates and badges with O-RAN logos 2. Removing the sharing mechanism in Chapter 5 related to self-certification by vendors, individual operators, and 3rd party testing houses/labs 3. Adding in Chapter 5 the requirement on universal machine-readable format in test data sharing 4. Making test case 4.2.4.6 in Chapter 7 mandatory 5. Modifying the wording in Section 7.2.1 to be more concise and refer to Fronthaul IOT test specification clearly 6. Revising Figure 5.2-1 and Table 6.2-1 to sync up with the latest reference documents 7. In the companion test report template, making the following changes • Adding tab for O-DU Conformance • Adding new test cases defined in WG4 CONF v04.00 to "FH O-RU Conformance" tab • Editorial corrections throughout the "FH O-RU Conformance" tab
2022.03.22	04.00	1. Adding section 5.3 on available certificates and badges to be issued under this release 2. Adding section 5.1.4 on the composition of O-RAN certificates and badges; Revising section 5.2.8.2 and 5.2.8.4 on test result sharing location and format; Revising the Certificate/Badge reference identifier 3. Revising section 5.2.5 and 5.2.6 on verifying test results and issuing certificates/badges; Revising section 4.3.2 to clarify what equipment can be awarded O-RAN certificate/badges 4. Revising section 6.4.2 to exclude LTE tests in various scenarios when O-RU does not support LTE 5. Many corrections and clean-ups in FH O-RU Conformance Tab inside the companion test report template 6. Many IOT updates in Chapter 7 to align with Nov-2021 WG test specifications 7. Adding all corresponding new test cases in various IOT TABs inside the companion test report template 8. Automating reference version updates inside the companion test report template by adding a "References" tab to list all documents that are referenced on other tabs
2022.07.27	05.00	1. Decoupling of O-RAN certification and badging processes and procedures from definition of tests in WG/FG test specifications – chapters 6, 7 and 8 2. Revising section 5.2.8.2 to provide further clarifications on centralized O-RAN web catalogue for issued O-RAN certificates and badges
2022.11.10	06.00	1. Adding Summary report (O-RAN Certificate and Badge Summary Report) and Test report (O-RAN Certificate and Badge Test Report), and updating whole document with these new terms 2. Updating section "7.2 O-RAN Open Fronthaul interface between O-RU and O-DU" with the terminology used in the latest version of O-RAN.WG4.IOT. 3. Adding section "5.2.10 Multi-vendor Scenario" introducing multi-vendor scenario to O-RAN certification and badging processes
2023.03.24	07.00	1. Clarification of available certificates and badges (Chapters 5.3, 6, 7, 8) 2. Adding subsystem combined O-DU/O-CU to OpenFH 3. Clarification of the tests which need to be passed (conducted) – Chapters 6, 7, 8 4. Updated process in Chapter 5.2 • Clarification of issuing of certificates and badges • Updated processes for storing and sharing of summary report, test reports and measurement results.
2023.11.09	08.00	1. Removing references not quoted 2. Clarification on the relocation of end-to-end security test cases to WG11 Security Test Specification

2024.03.22	09.00	1. Clarification on F1/Xn IOT badge by including both user-plane and control-plane test cases based on the latest updates in O-RAN.WG5.IOT 2. Clarification on OpenFH IOT badge based on the latest updates in O-RAN.WG4.IOT related to new assigned status (M/CM/O) for S-plane IOT test cases
2024.07.18	10.00	1. Updated requirements on the summary and test reports to improve confidentiality – the reports shall be encrypted and digitally signed. 2. To ensure reusability and machine-readability of summary report, the unified file format (.xls/.xlsx) shall be used.
2025.03.21	11.00	1. Added the new process to enable awarding of O-RAN certificates/badges for products from non-O-RAN member entities 2. Introduced machine readable format for summary report, test report and measurement results 3. Updated O-RAN Cert/Badging process with the following • O-RAN Certification and Badging process flow diagram • Option to define O-RAN Certification and Badging Profile • New O-RAN Certification and Badging ICS/ICS Proforma
2025.07.18	12.00	1. Added the new process to enable the product owner (while not the original product vendor/manufacture) to request for O-RAN cert/badging 2. Introduced O-RAN certificates and badges for A1 services on A1 interface 3. Clarified O-RAN CB-Profiles and CB-ICS Proformas as two new subsidiary documents to the main Cert/badging process and procedure document 4. Defined the detail O-RAN CB-Profile and CB-ICS Proforma for A1 services 5. Updated the OpenFH IOT badging process corresponding to the newly added "to be validated" list for some PTCs in the latest O-RAN.WG4.TS.IOT.0-R004-v13.00

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